42.6: Breathing ventilates the lungs

Like fishes, terrestrial vertebrates rely on ventilation to maintain high O_2 and low CO_2 concentrations at the gas exchange surface. The process that ventilates lungs is **breathing**, the alternating inhalation and exhalation of air. A variety of mechanisms for moving air in and out of lungs have evolved, as we will see by considering breathing in amphibians, birds, and mammals.

How an Amphibian Breathes

An amphibian such as a frog ventilates its lungs by **positive pressure breathing**, inflating the lungs with forced airflow. Inhalation begins when muscles lower the floor of an amphibian's oral cavity, drawing in air through its nostrils. Next, with the nostrils and mouth closed, the floor of the oral cavity rises, forcing air down the trachea. Exhalation follows as air is expelled by the elastic recoil of the lungs and by compression of the muscular body wall. When male frogs puff themselves up in aggressive or courtship displays, they disrupt this breathing cycle, taking in air several times without allowing any release.

How a Bird Breathes

When a bird breathes, it passes air over the gas exchange surface in only one direction. Air sacs situated on either side of the lungs act as bellows that direct air flow through the lungs. Within the lungs, tiny channels called *parabronchi* serve as the sites of gas exchange. Passage of air through the entire system—air sacs and lungs—requires two cycles of inhalation and exhalation **(Figure 42.26)**.

Figure 42.26

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The avian respiratory system.
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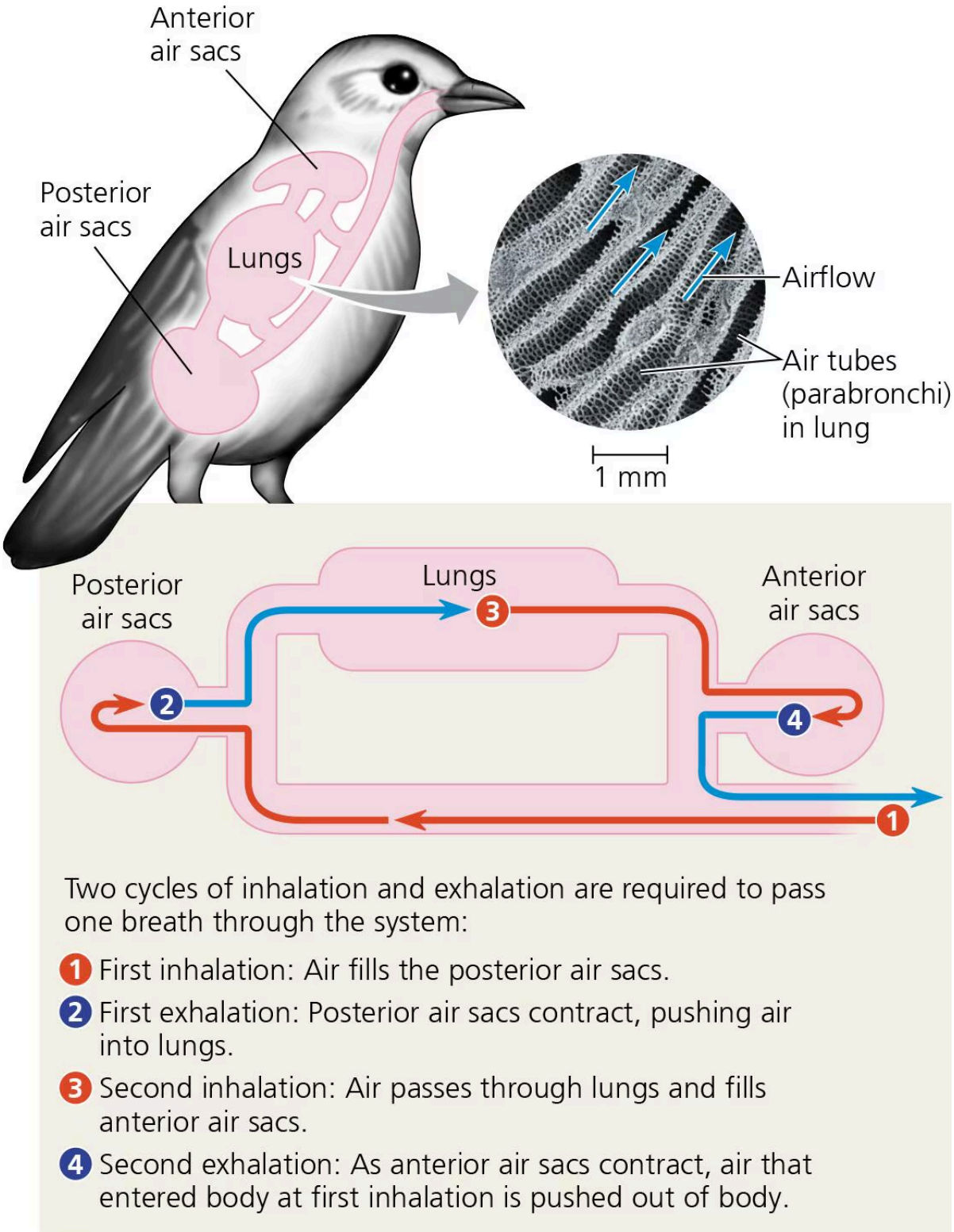

This diagram traces a breath of air through the respiratory system of a bird.

Diagram of the avian respiratory system shows a bird with anterior air sacs, followed by the lungs and then the posterior air sacs. Within the lungs, air flows only in one direction through the air tubes (parabronchi) in the lungs. Parabronchi appear as rows of ridges in an image (scale: 1 micrometer). Two cycles of inhalation

and exhalation are required to pass one breath through the system as follows:

1. First inhalation: Air fills the posterior air sacs.
2. First exhalation: Posterior air sacs contract, pushing air into lungs.
3. Second inhalation: Air passes through lungs and fills anterior air sacs.
4. Second exhalation: As anterior air sacs contract, air that entered body at first inhalation is pushed out of body.

Ventilation in birds is highly efficient. One reason is that birds pass air over the gas exchange surface in only one direction during breathing. In addition, incoming fresh air does not mix with air that has already carried out gas exchange, maximizing the partial pressure difference with blood flowing through the lungs.

How a Mammal Breathes

To understand how a mammal breathes, think about filling a syringe. By pulling back on the plunger, you lower the pressure in the syringe, drawing gas or fluid through the needle into the syringe chamber. Similarly, mammals employ **negative pressure breathing**—pulling, rather than pushing, air into their lungs **(Figure 42.27)**. Using muscle contraction to actively expand the thoracic cavity, mammals lower air pressure in their lungs below that of the air outside their body. Because gas flows from a region of higher pressure to a region of lower pressure, the lowered air pressure in the lungs causes air to rush through the nostrils and mouth and down the breathing tubes to the alveoli.

Figure 42.27

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Negative pressure breathing.
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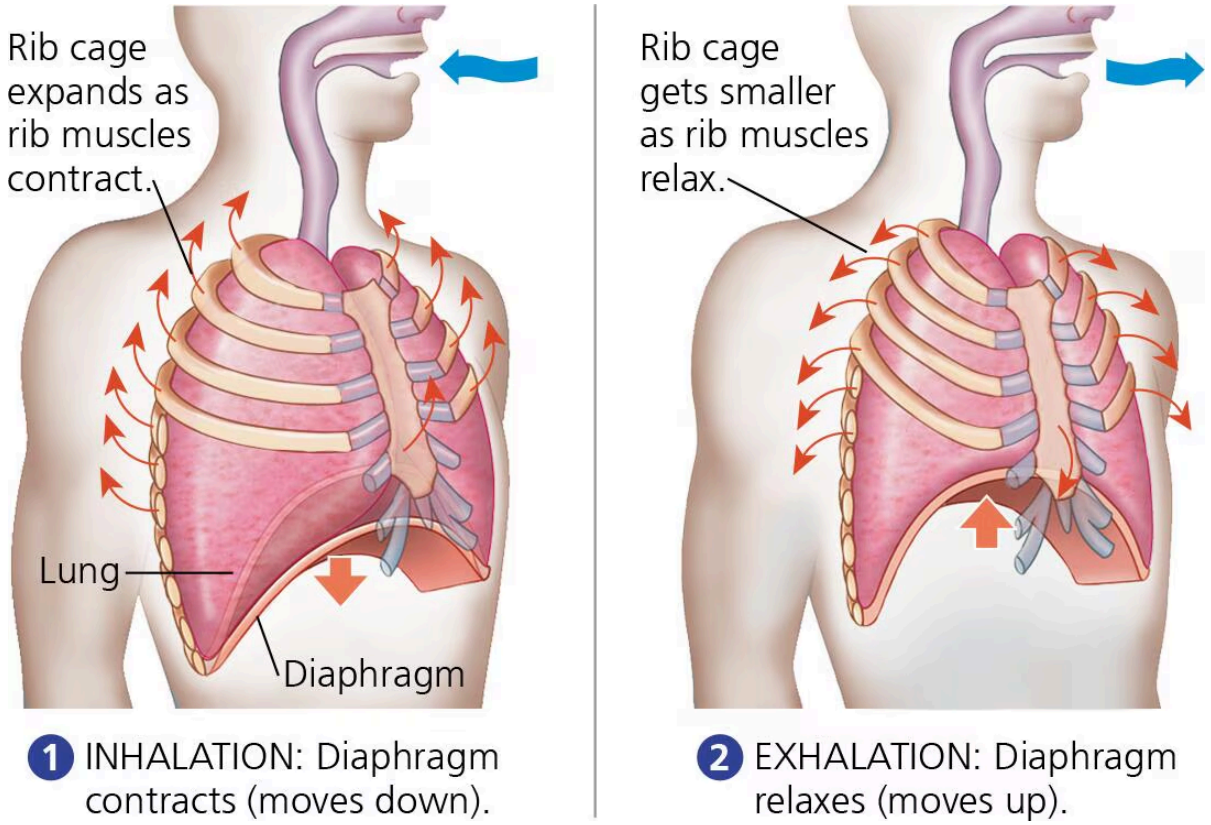


A mammal breathes by changing the air pressure within its lungs relative to the pressure of the outside atmosphere.

Negative pressure breathing in a human (mammal).

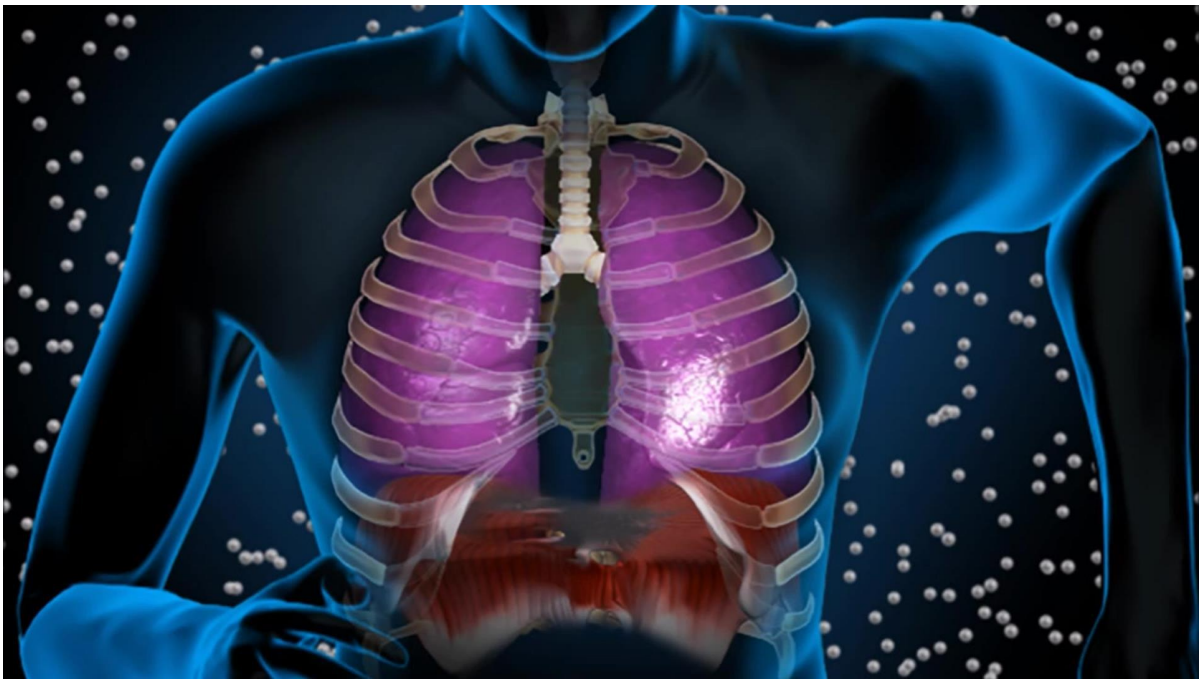
1. Inhalation: Diaphragm contracts (moves down). Rib cage expands as muscles contract. Air moves in through mouth.
2. Exhalation: Diaphragm relaxes (moves up). Rib cage gets smaller as rib muscles relax. Air moves out through mouth.

WHAT IF? *The walls of alveoli contain elastic fibers that allow the alveoli to expand and contract with each breath. If the alveoli lost their elasticity, how would that affect gas exchange in the lungs?*

*For suggested answer, see*Appendix A.*

BioFlix® Animation

Mechanics of Breathing



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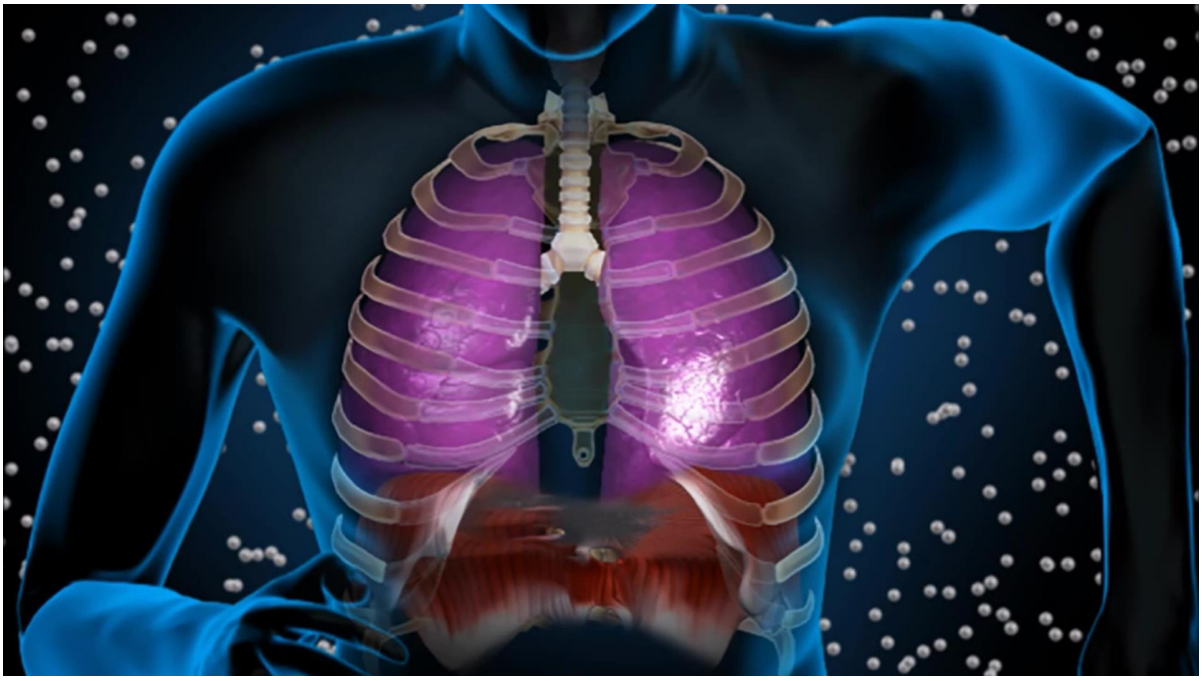

Expanding the thoracic cavity during inhalation involves the animal's rib muscles and the **diaphragm**, a sheet of skeletal muscle that forms the bottom wall of the cavity. Contracting the rib muscles pulls the ribs upward and the sternum outward, expanding the rib cage, the front wall of the thoracic cavity. At the same time, the diaphragm contracts, expanding the thoracic cavity downward. It is this descending movement of the diaphragm that is analogous to a plunger being drawn out of a syringe.

Whereas inhalation is always active and requires work, exhalation is usually passive. During exhalation, the muscles controlling the thoracic cavity relax, and the volume of the cavity is reduced. The increased air pressure in the alveoli forces air up the breathing tubes and out of the body.

Within the thoracic cavity, a double membrane surrounds the lungs. The inner layer of this membrane adheres to the outside of the lungs, and the outer layer adheres to the wall of the thoracic cavity. A thin space filled with fluid separates the two layers. Surface tension in the fluid causes the two layers to stick together like two plates of glass separated by a film of water: The layers can slide smoothly past each other, but they cannot be pulled apart easily. Consequently, the volume of the thoracic cavity and the volume of the lungs change in unison.

The rib muscles and diaphragm are sufficient to change lung volume when a mammal is at rest. During exercise, other muscles of the neck, back, and chest increase the volume of the thoracic cavity by raising the rib cage. In kangaroos and some other mammals, locomotion causes a rhythmic movement of organs in the abdomen, including the stomach and liver. The result is a piston-like pumping motion that pushes and pulls on the diaphragm, further increasing the volume of air moved in and out of the lungs.

The volume of air inhaled and exhaled with each breath, called **tidal volume**, averages about 500 mL in resting humans. The tidal volume during maximal inhalation and exhalation is the **vital capacity**, about 3.4 L and 4.8 L for college-age women and men, respectively. The air that remains after a forced exhalation is called the **residual volume**. With age, the lungs lose their resilience, and residual volume increases at the expense of vital capacity.

Because the lungs in mammals do not completely empty with each breath, and because inhalation occurs through the same airways as exhalation, each inhalation mixes fresh air with oxygen-depleted residual air. As a result, the maximum PO_2 in alveoli is always considerably less than in the atmosphere. The maximum PO_2 in lungs is also less for mammals than for birds, which have a unidirectional flow of air through the lungs. This is one reason why mammals function less well than birds at high altitude. For example, humans have great difficulty obtaining enough O_2 when climbing at high elevations, such as those in the Himalayas. However, bar-headed geese and several other bird species easily fly through high Himalayan passes during their migrations.

CheckPoint

Ventilation and Respiration

Check your understanding before you move on, with the CheckPoint: Ventilation and Respiration.

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Control of Breathing in Humans

Although you can voluntarily hold your breath or breathe faster and deeper, most of the time your breathing is regulated by involuntary mechanisms. These control mechanisms ensure that gas exchange is coordinated with blood circulation and with metabolic demand.

The neurons mainly responsible for regulating breathing are in the medulla oblongata, near the base of the brain **(Figure 42.28)**. Neural circuits in the medulla form a pair of *breathing control centers* that establish the breathing rhythm.

Figure 42.28

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Homeostatic control of breathing.
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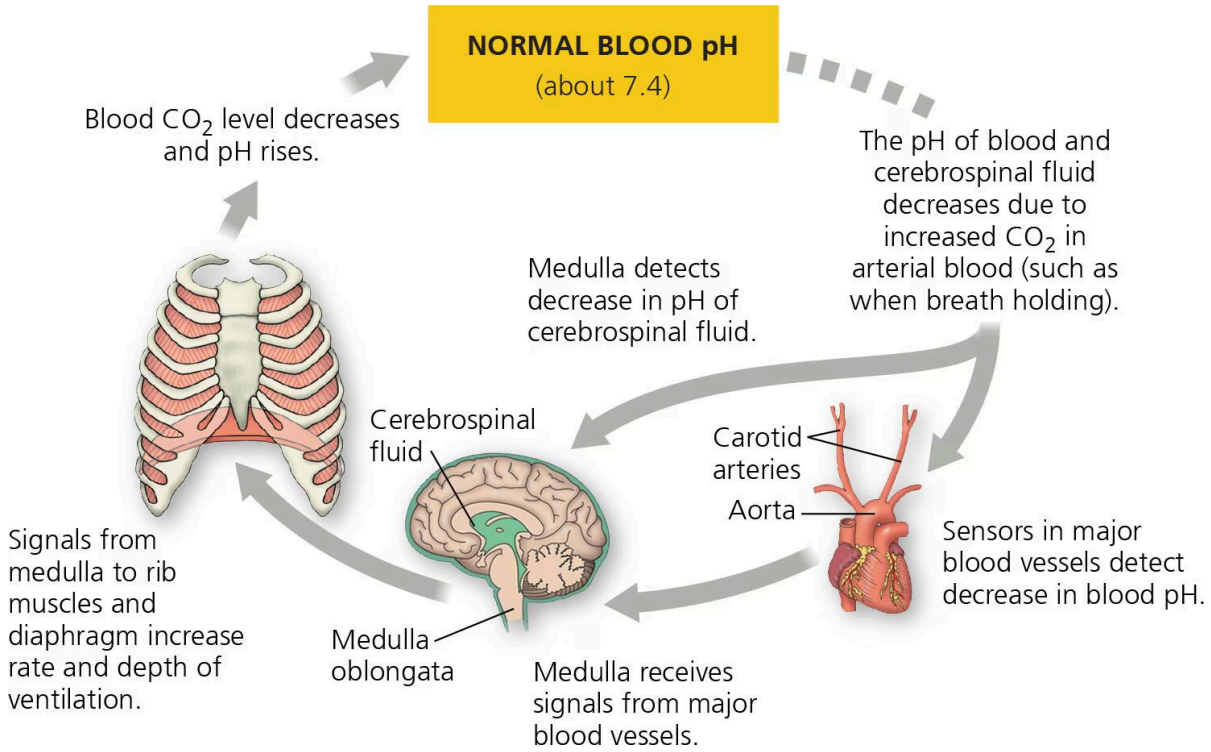

Flow diagram depicts control of breathing. Normal blood p H is about 7.4. Blood p H falls due to rising levels of C O 2 in tissues (such as when exercising). Two results: (1) Medulla detects decrease in p H of cerebrospinal fluid and (2) Sensors in major blood vessels (aorta, carotid arteries) detect decrease in blood p H. The medulla oblongata (base of brain) receives signals from the major blood vessels as well as the cerebrospinal fluid. Signals from the medulla to rib muscles and diaphragm increase the rate and depth of ventilation. Blood C O 2 level falls and p H rises, returning the blood to normal p H of about 7.4.

VISUAL SKILLS *Suppose a person began breathing very rapidly while resting. Tracing a path along this negative-feedback control circuit, describe the effect on the* CO_2 *level in the blood and the steps by which homeostasis would be restored.*

For suggested answer, see *Appendix A*.*

In regulating breathing, the medulla uses the pH of the fluid in which it is bathed as an indicator of blood CO_2 concentration. The pH can be used in this way because blood CO_2 is the main determinant of the pH of cerebrospinal fluid, the fluid surrounding the brain and spinal cord. Carbon dioxide diffuses from the blood to the cerebrospinal fluid, where it reacts with water and forms carbonic acid (H_2CO_3). The H_2CO_3 can then dissociate into a bicarbonate ion (HCO_3^-) and a hydrogen ion H^+:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

Consider what happens when you hold your breath, and CO_2 accumulates in your blood and cerebrospinal fluid. Through the reactions shown above, the higher CO_2 concentration leads to an increase in the concentration of H^+, lowering pH. Sensors in the medulla as well as in major blood vessels detect this pH change. In response, the medulla's control circuits increase the depth and rate of breathing (see Figure 42.28). Both remain high until the excess CO_2 is eliminated in exhaled air and pH returns to a normal value.

The blood O_2 level usually has little effect on the breathing control centers. However, when the O_2 level drops very low (at high altitudes, for instance), O_2 sensors in the aorta and the carotid arteries in the neck send signals to the breathing control centers, which respond by increasing the breathing rate and depth. The regulation of breathing is modulated by additional neural circuits, primarily in the pons, a part of the brain next to the medulla.

Breathing control is effective only if ventilation is matched to blood flow through alveolar capillaries. During exercise, for instance, such coordination couples an increased breathing rate and depth, which enhance O_2 uptake and CO_2 removal, with an increase in cardiac output. When you breathe deeply, a negative-feedback mechanism prevents the lungs from overexpanding: During inhalation, sensors that detect stretching of the lung tissue send nerve impulses to the control circuits in the medulla, inhibiting further inhalation.